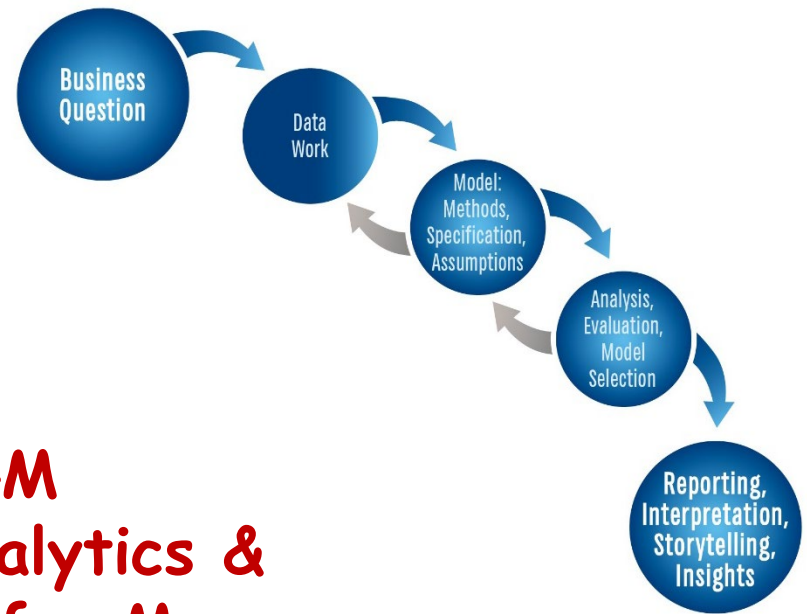




PAML4M

Predictive Analytics & Machine Learning for Managers

2. Foundations

Script: PAML4M_2R_Foundations.Qmd (Github)

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Last updated 8/1/2026



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ITEC 621

Predictive Analytics

2.1 Stats Refresher



Analytics Modeling Options

	Modeling Method		
	Structured	Visual, Text, Unstructured, etc.	
Descriptive	Cluster analysis, correlation, market basket analysis, sample statistics, ANOVA		Bubble charts, network diagrams, natural language processing, clustering dendograms, etc.
Predictive	Association	Decision Tree	Charts
Quantitative Value	Regression	Regression Trees	Regression plots, scatter plots, Tableau diagrams, trend charts, etc.
Classification	Logistic Regression; Other Categorical Regression Models	Classification Trees	Tree maps, interactive diagrams,
Prescriptive	Operations research, decision modeling, optimization, linear programming		Simulations, etc.

Descriptive Analytics

Understanding Variability

- **Variance**

- Explains how spread out are the values of a **single variable** from its mean

- **Covariance**

- Explains how **2 variables** co-vary together
- Covariance can be **difficult to evaluate** if the 2 variables are of **different types** (e.g., categorical x quantitative) or have **dissimilar scale** (e.g., miles per gallon vs. kilometers per liter will yield different values)

Variance Refresher

- A lot of what we do in analytics is understanding and explaining variance and covariance. **Knowing these concepts is key**
- **Variance** is how much values vary relative to the mean. The value is squared so that values, both below and above the mean contribute positively to the variance statistic
- The square root of the variance is the **standard deviation**

$$Var(x) = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1} = \sigma(x)^2 = Std Dev(x)^2$$

Covariance and Correlation

Covariance Challenges

Because covariance is affected by **scale** and **data types**, we often use the following instead:

- **Quantitative** x **Quantitative** → **Correlation** → normalize the variables so that the correlation range is from -1 to 1
- **Quantitative** x **Categorical** → **ANOVA** → analyze if the means vary widely across categories
- **Categorical** x **Categorical** → χ^2 (Chi Square) **Test of Independence** → evaluate if values of one categorical variable values are associated with values from another (dependent) or not (independent)

Covariance Refresher

- **Covariance** indicates how two **quantitative** variables are related – i.e., how do they “**co-vary**” or how they move together. If when x is above (or below) the mean y is generally above (or below) the mean, then the covariance is positive. Otherwise, is negative.

$$\text{Cov}(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

- Covariance is a useful concept, but it has limited practical use because the covariance value is dependent on the **scale** used to measure x and y .
- **Correlation** takes care of this scale problem.

Correlation Refresher (Quant x Quant)

- Correlation measures **co-variability** like covariance, but the deviation from the mean of each variable are **divided by the standard deviation** of the variables → it can be shown mathematically that the correlation of two variables will not change with **re-scaling**.
- Mathematically, correlation statistics ranges from **-1.0** to **1.0**
- Provides an indication for whether two **quantitative** variables vary in the same or opposite direction, or not (**independent**)

$$\rho(x, y) = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{\sigma_x} \right) \left(\frac{y_i - \bar{y}}{\sigma_y} \right) = \frac{Cov(x, y)}{\sigma_x \sigma_y}$$

- Analyzing the **descriptive statistics** (i.e., mean and standard deviation) and the **correlation** among all variables is a necessary **first step** (i.e., descriptive analytics) before building predictive models → it is good to **eye-ball** the data → **Ideally**:

- ✓ **Predictors** should be **correlated** with the **outcome** variable
- ✓ But should be somewhat **uncorrelated** with **each other**

2 Key Correlation Values

1. **Magnitude** – how large is the association

$\rho = 1.0$ Perfectly positively correlated

$\rho = +$ Positively correlated

$\rho =$ Around 0 Uncorrelated (i.e., independent)

$\rho = -$ Negatively correlated

$\rho = - 1.0$ Perfectly negatively correlated

2. **Significance** – probability that the observed correlation happened by chance – i.e., **$H_0: \rho=0$** (no correlation)

$p \geq 0.10 \rightarrow$ Not significantly $\neq 0 \rightarrow$ i.e., independent

$p < 0.10 \rightarrow$ Moderately significant

$p < 0.05 \rightarrow$ Significant

$p < 0.01 \rightarrow$ Very significant

$p < 0.001 \rightarrow$ Highly significant

It is useful to inspect the **correlation matrix** $\rightarrow$



Examples: Correlation

- {library} Topic
Search keywords
- {Hmisc} Correlation matrix with p-values
`rcorr(as.matrix(mtcars))`
- {mtCorr} `corrplot( )` Correlation graphs
`mtCorr <- cor(mtcars)`



ANOVA

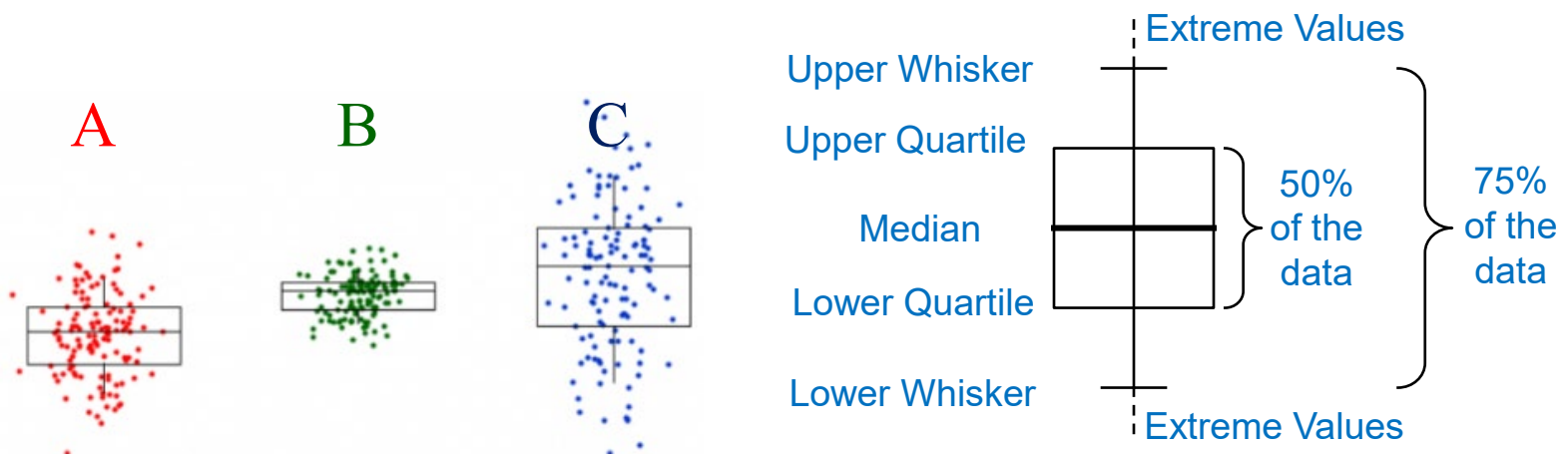
(Quant x Categorical)

ANOVA Refresher

- **Analysis of variance** (ANOVA) tests whether the mean of a given **quantitative** variable is the same between two or more **categories** or groups.
- It is called analysis of “**variance**” and not analysis of “**means**” because it compares the variance within each group against the variance of the means between groups.
- For example, if we want to test if the mileage is different between foreign and domestic cars, or if the price of a diamond is different for various color classifications, we can do an ANOVA test.
- The **intuition** is that if the variance **between groups** is significantly larger than the variance **within groups**, then the means are significantly different across groups. Otherwise, they are not.

Boxplots: ANOVA Visual Exploration

- Boxplots are excellent to explore variation of means across categories
- In the boxplot below, you can inspect visually how the data varies around the mean for each category **A**, **B** and **C**.
- If the boxplot means and boxes roughly **align**, **ANOVA** will most likely yield a **non-significant** p-value
 - i.e., H_0 : No difference in means across categories → **supported !!**
- If the means and boxes in some categories are noticeably **higher** or **lower** than others, **ANOVA** will most likely yield a significant p-value
 - i.e., H_0 : No difference in means across categories → **rejected !!**



How to Read a Boxplot

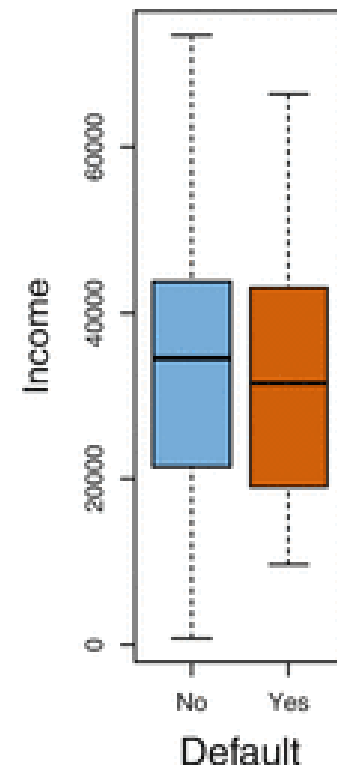
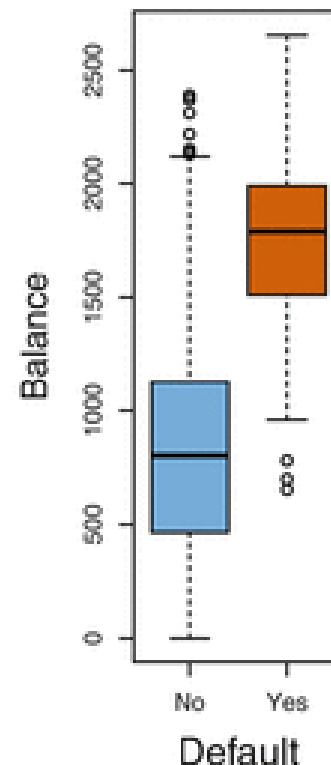
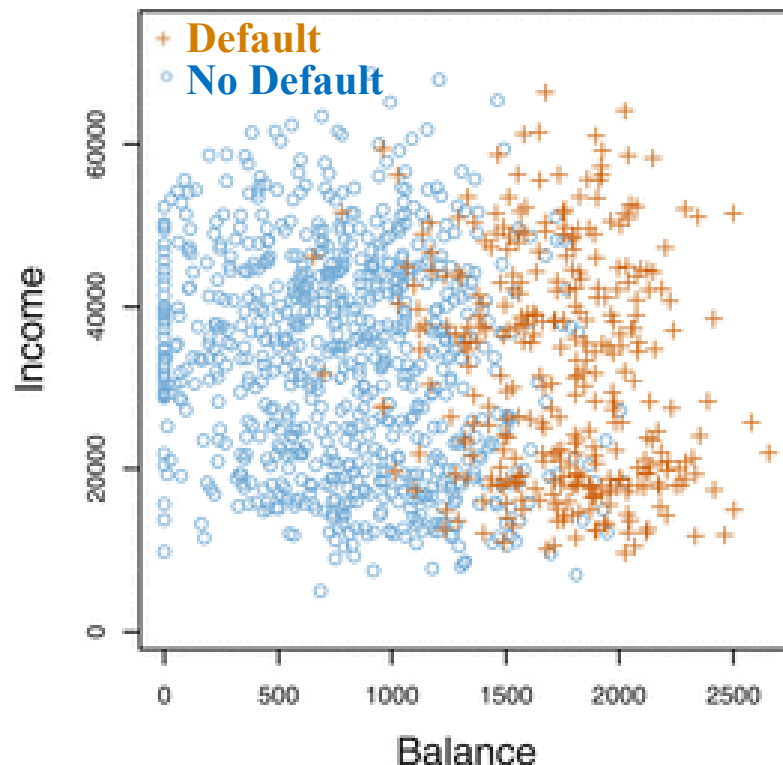
Illustration: Eyeballing ANOVA

```
aov(Balance ~ Default, data=Credit)
```

→ Probably significant → means & boxes very different

```
aov(Income ~ Default, data=Credit)
```

→ Probably not significant → means & boxes align





Examples: ANOVA

- `{library}` Topic
Search keywords
- `{ggplot2}`(diamonds) ANOVA Summary Statistics
`summary(aov(price ~ clarity,`
- `{ggplot2}` (diamonds) Box Plots
`boxplot(price ~ clarity,`

Chi Square Test of Independence

(Categorical $\times$ Categorical; Non-Parametric)

Chi Square Test of Independence

- Suppose that you want to test the effectiveness of two donation campaigns A and B, on driving donations
- **Correlation** and **ANOVA** will **NOT** work to evaluate if one campaign was more successful than the other because Campaign and Donation are categorical variables
- The most common way to test whether two categorical variables are **related** (one **influences** the other) or **independent** (**no influence**) is with a **Chi Square Test of Independence**
- The first step is to **cross-tabulate** the data with Donation (yes or no) as rows and Campaign (A or B) as columns and enter the respective counts in the cells.
- Then run a **Chi Square** test on the **cross table**

Donor	Campaign	Donation
1	A	Yes
2	A	Yes
3	B	No
4	B	Yes
5	A	No
Etc.	B	Yes

Chi Square Test Explained

Campaign has no effect

	Campaign		
Donation	A	B	Total
No	2,820 (2,800)	4,180 (4,200)	7,000
Yes	1,180 (1,200)	1,820 (1,800)	3,000
Total	4,000	6,000	10,000

Campaign has effect

	Campaign		
Donation	A	B	Total
No	2,100 (2,800)	4,900 (4,200)	7,000
Yes	1,900 (1,200)	1,100 (1,800)	3,000
Total	4,000	6,000	10,000

- **Observed** (actual) counts
- **Expected** by **chance** if Campaign and Donation were independent

$$= \frac{7,000 * 4,000}{10,000} = 2,800$$

- The **Chi Square Test of Independence** tests if the **difference** between **observed** (actual) counts and **expected** counts by chance (if Donation and Campaign are independent) is significant

- ✓ If **not significant** → **retain** H_0 : Campaign and Donation are **independent**
- ✓ If **significant** → **reject** H_0 : Campaign **influences** Donation

Illustration

Margin Totals

COVID	Test A	Test B	Total
Positive			100
Negative			900
Total	400	600	1000

Expected by Chance (random)

Diagnosis is **independent** from test

COVID	Test A	Test B	Total
Positive	40	60	100
Negative	360	540	900
Total	400	600	1000

Observed Case 1: Diagnosis depends on Test

A yields **more positives** and less negatives

COVID	Test A	Test B	Total
Positive	80	20	100
Negative	320	580	900
Total	400	600	1000

Difference btw observed and chance

COVID	Test A	Test B	Sum ²
Positive	40	-40	3200
Negative	-40	40	3200
Sum ²	3200	3200	6400

Person	Test	Diagnosis
John	A	Positive
Sally	A	Negative
Sue	B	Negative
Bob	A	Positive
Al	B	Negative
Mary	B	Negative
Etc.		

Observed Case 2: Diagnosis depends on Test

A yields less positives and **more negatives**

COVID	Test A	Test B	Total
Positive	10	90	100
Negative	390	510	900
Total	400	600	1000

Difference btw observed and chance

COVID	Test A	Test B	Sum ²
Positive	-30	30	1800
Negative	30	-30	1800
Sum ²	1800	1800	3600



Chi-Square Test of Independence

- `{library}` Topic
Search keywords
- `{ggplot2}`(diamonds) Chi-Square Test Summary Statistics
`cross.table <- table (cut, color)`
`chiSq.diamonds <- chisq.test(cross.table)`

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Predictive Analytics

2.2 Regression Refresher

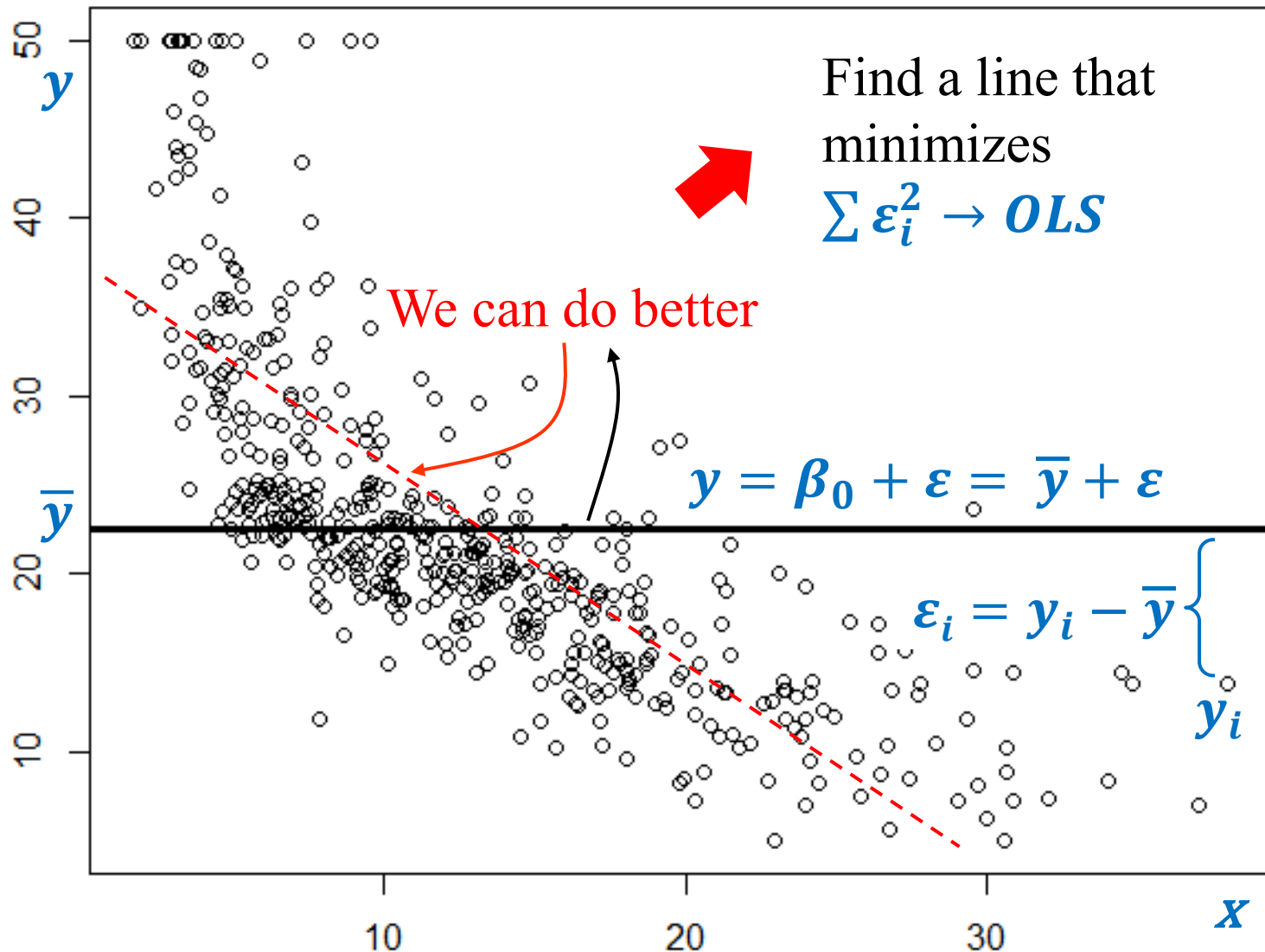


Analytics Modeling Options

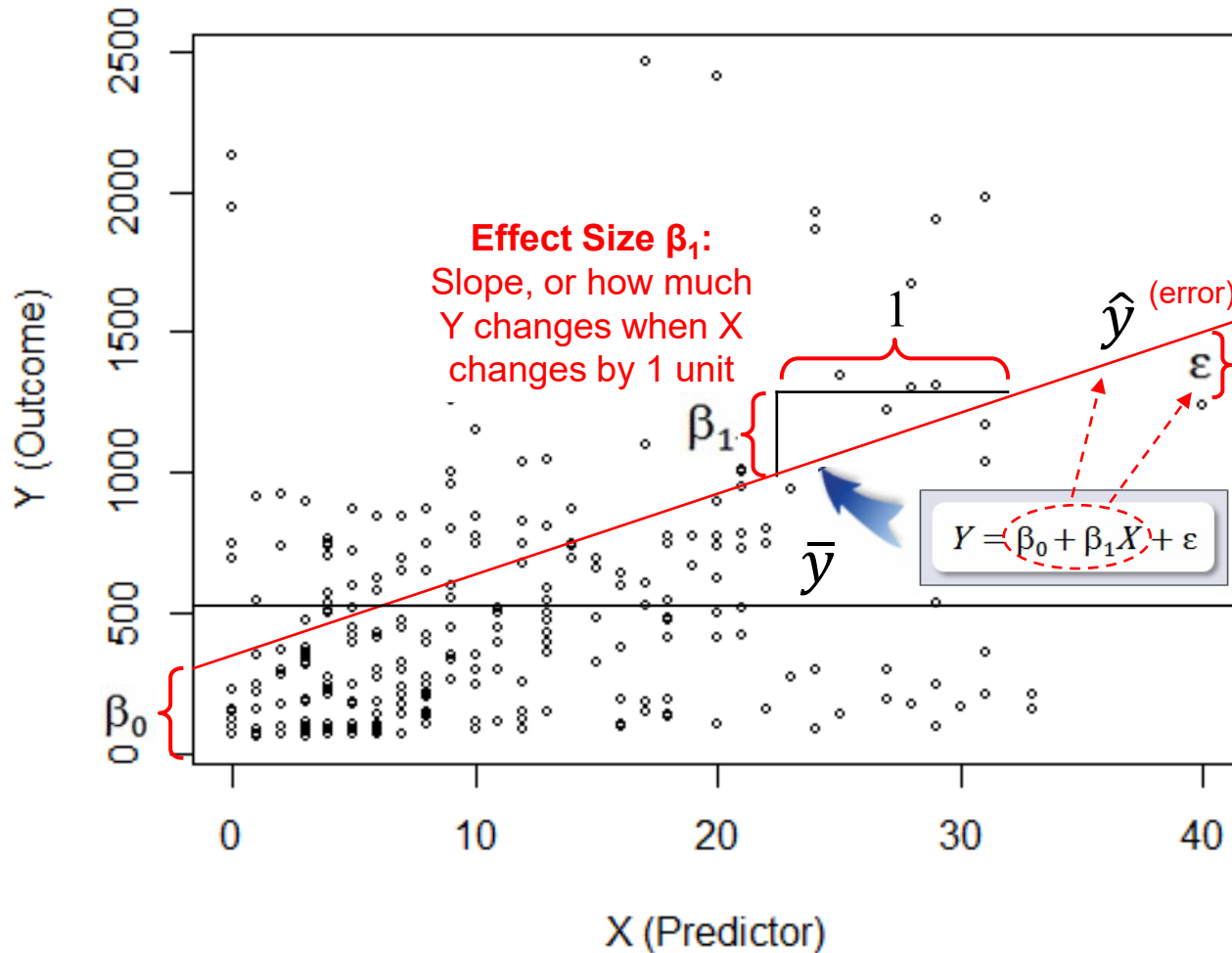
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Prescriptive	Operations research, decision modeling, optimization, linear programming		Simulations, etc.

Simple Linear Regression

The Null Model: No Predictors



Simple Linear Regression

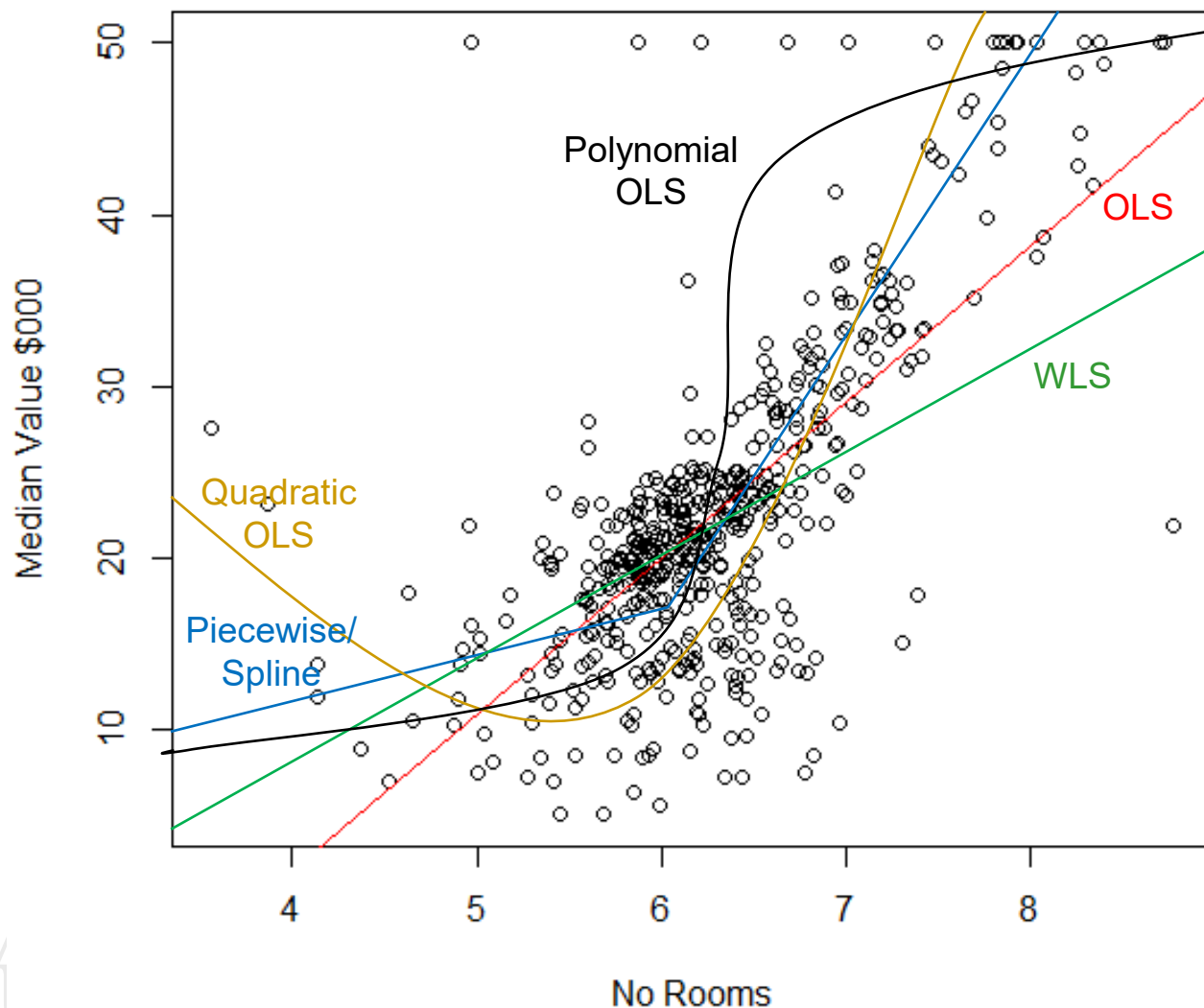


Questions

- How do you find the regression **line** that best **fits** your **data**?
- How do you **minimize errors** (i.e., deviations from the regression line)?
- In statistical parlance, how do you “**estimate**” (**fit**) the regression line?
- There are many “**estimation methods**”
- The most popular one is called “**Ordinary Least Squares**” (**OLS**)
- It is the line that **minimizes the sum of errors squared**

Estimation (Boston{MASS})

Regression for Boston House Values



Questions

- **Estimation** is about the **method** and **model specification** used to train your model
- Is **OLS** the best estimation method?
- **OLS** minimizes the sum of squared errors.
- But weighted least squares (**WLS**) may be more efficient method when residuals are uneven
- **Quadratic OLS**
- **Piecewise**, or **Spline** regressions may be more accurate predictors
- Polynomial regressions may even be more accurate but it suffers from “over-identification”.

**Selecting your
model method
is key!!**

Important Regression Statistics

- β_0 = The **intercept** $\rightarrow$ Y's predicted value when $X = 0$
- β_1 = Slope of the regression line – i.e., change in Y when X increases by 1 unit or the “**effect size**”
- **Significance** = whether β_1 is significantly **different than 0**
- **p-value** = measures **significance** (same as with correlation) i.e., the probability that $H_0: \beta_0, \beta_1, \text{etc.} = 0$
 $\rightarrow$ i.e., the observed effects happened by chance
 $\rightarrow$ the **smaller** the **p-value** the **better**
- **Confidence** = **1 - p**; e.g., $p < 0.05$ implies we are more than (0.95 or) 95% confident that the effect did not happen by chance
- **Standard Errors (SE)**: useful to eyeball results $\rightarrow$ the 95% confidence interval for a coefficient is $\beta \pm 2*SE$

OLS Model Diagnostics



OLS: Fit Statistics

- y_i = *Actual*; $\hat{y}_i$ = *Predicted* (regression line); $\bar{y}$ = *Mean*
- Total sum of squares (*SST*) $\rightarrow \sum (y_i - \bar{y})^2$ (*null model error*)
- Regression sum of squares (*SSR*) $\rightarrow \sum (\hat{y}_i - \bar{y})^2$ (*predicted*)
- Sum of squared errors (*SSE*) $\rightarrow \sum (y_i - \hat{y}_i)^2 = SST - SSR$ – i.e., deviations from the regression line; i.e., part of *SST* unexplained by the regression line
- Degrees of freedom (*df*): the number of observations in a sample minus the number of parameters that restrict the data
- n = number of observations $\rightarrow df_{Total} = n - 1$
- p = number of predictors in the model $\rightarrow df_{Regression} = p$
- $df_{Residual} = df_{Total} - df_{Regression} = n - p - 1$
- Mean Squares: $MST = \frac{SST}{n}$ $MSR = \frac{SSR}{p}$ $MSE = \frac{SSE}{n}$
- Root Mean Square: $RMS = \sqrt{MSE}$



$$MSE = \frac{SSE}{n}$$

OLS Predictive Power and Fit

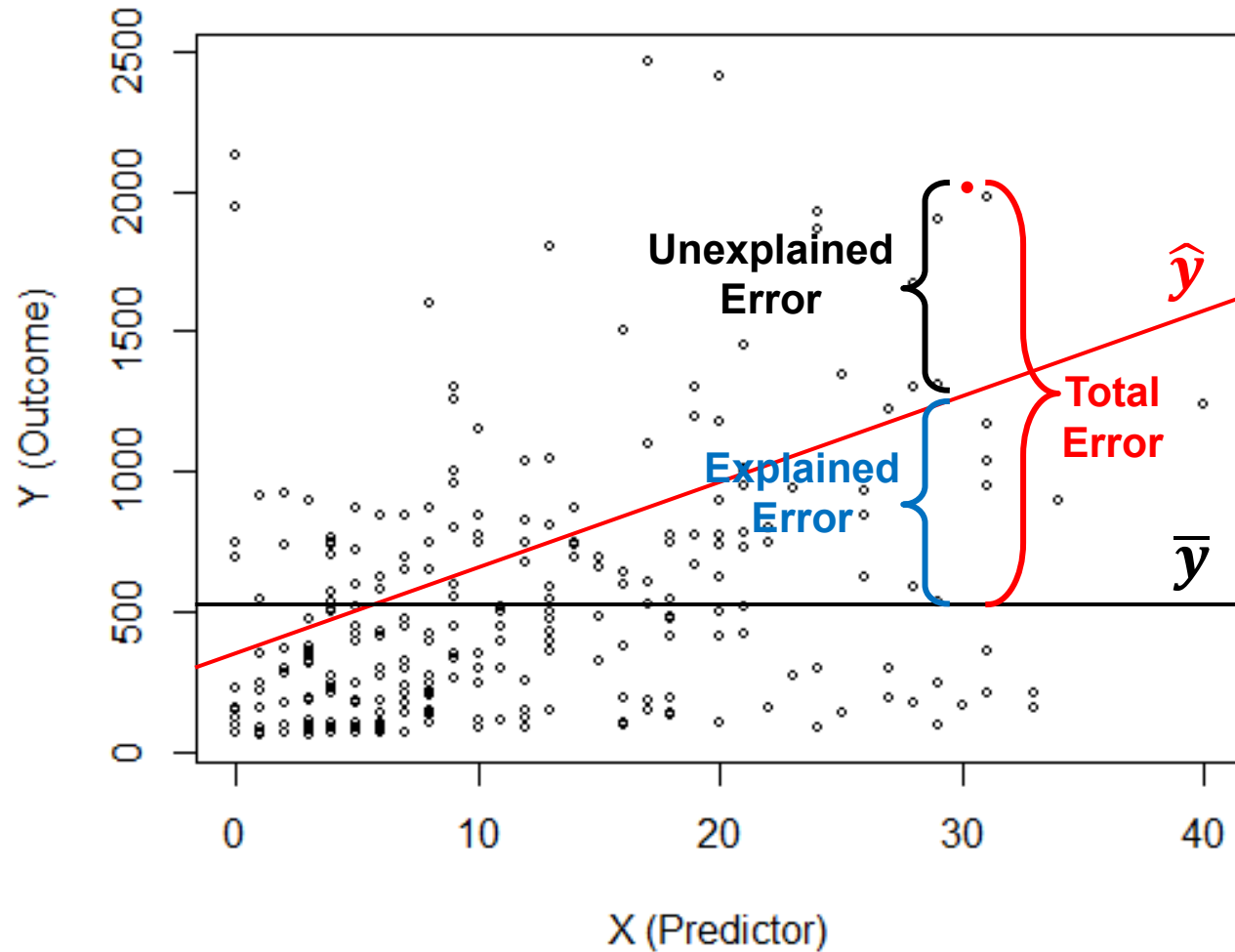
- R^2 (coefficient of determination) = proportion of **variance** of Y (the predicted variable) **explained** by the regression line:

$$R^2 = \frac{SSR}{SST} = \frac{SST - SSE}{SST} = 1 - \left(\frac{SSE}{SST} \right)$$

- Closer to 1 → More **variance** is **explained** by the regression model
- Think of it as the **squared correlation statistic** for the whole model – in fact, for simple regression models R^2 is equal to $\rho(y, x)^2$
- **ANOVA** in a regression compares the error **variance explained** by the **regression line** compared to the **null model** (error variance explained by the mean of y) → it yields an **F statistic**, which translates into a **p-value** → i.e., if $p < 0.05$ or smaller → **the regression model has more predictive power than the null model !!**
- Each time we add **one variable** to a model the variance explained increases → i.e., **R^2 goes up**. Thus, comparing R^2 's between two models of different size is misleading. The **adjusted R^2** corrects for this.

- **Adjusted R^2** = $1 - \frac{(1 - R^2)(n - 1)}{n - p - 1}$ → penalized by p

Understanding R^2 Further



$$R^2 = \frac{\text{Explained Variance}}{\text{Total Variance}} = 1 - \frac{\text{Unexplained Variance}}{\text{Total Variance}}$$



Simple Linear Regression

- {library} Topic
Search keywords
- {MASS} Simple Linear Model

```
lm.fit.null <- lm(medv ~ 1,  
lm.fit <- lm(medv ~ lstat,
```
- Important Regression Plots

```
plot(medv ~ lstat,  
plot(lm.fit) # All 4 residual plots  
plot(lm.fit, which = 2) # Residual QQ-Plot
```

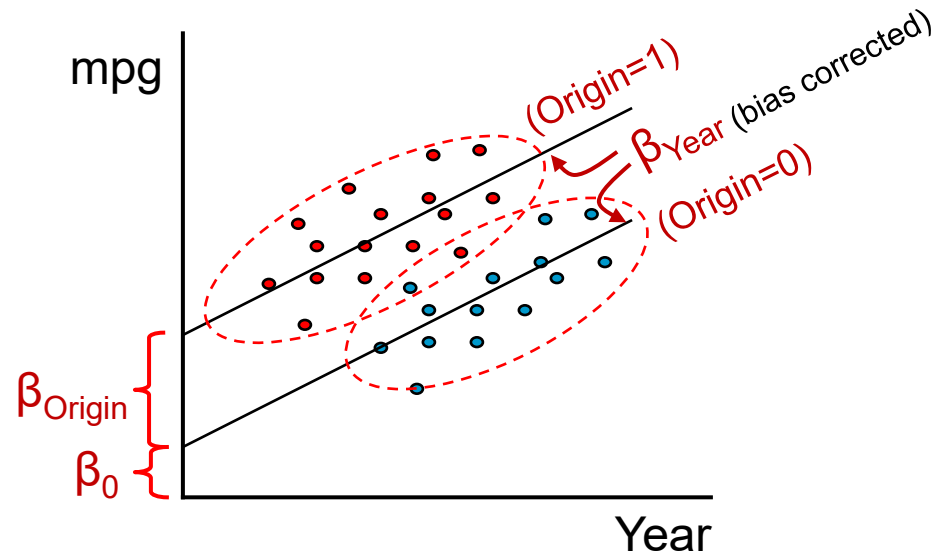
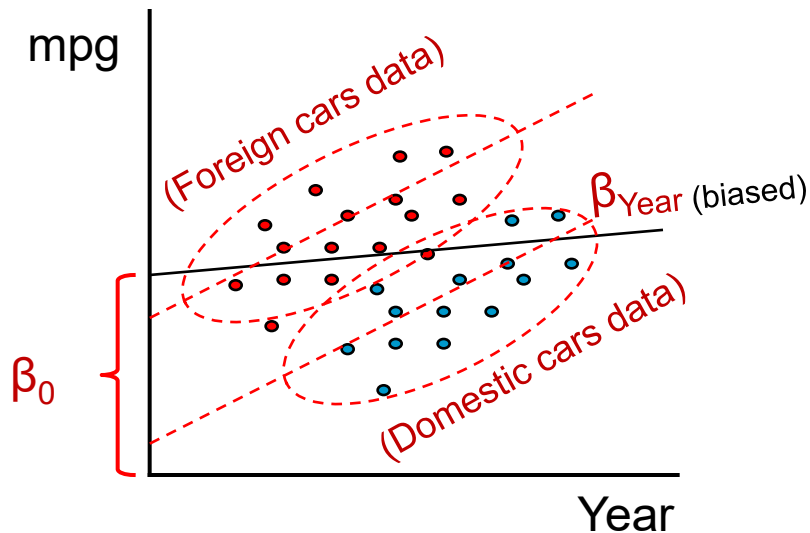
Dummy Variables

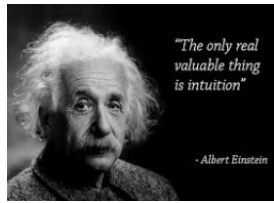
Adding a Dummy (Binary) Variable

- What if a car's year is a marginal predictor of mileage (mpg) but you suspect that this effect may be different between domestic and foreign cars?
- In this case, you can add a binary variable “*Origin*” to the model, which takes the value of *1* if the car is *foreign* and *0* if it is *domestic*.
- The model will then look like this (*Year* is continuous; *Origin* is 0 or 1):

$$\text{mpg} = \beta_0 + \beta_{\text{Year}}(\text{Year}) + \varepsilon$$

$$\text{mpg} = \beta_0 + \beta_{\text{Origin}}(\text{Origin}) + \beta_{\text{Year}}(\text{Year}) + \varepsilon$$





Dummy Variables: Intuition

- A regression with one or more dummy variables is very common in predictive modeling.
- As the diagram in the previous slide illustrates a dummy variable shows a parallel shift in the regression line. For example, a model that predicts miles per gallon in a vehicle based on the vehicle origin and its year is:

$$\text{mpg} = \beta_0 + \beta_{\text{Origin}}(\text{Origin}) + \beta_{\text{Year}}(\text{Year}) + \varepsilon$$

- $\text{Origin} = 0$ for **domestic** cars and $\text{Origin} = 1$ for **foreign** cars
- β_{Year} is the effect of vehicle year on mileage “controlling for” (taking into account) vehicle origin.
- β_{Origin} is the effect of origin on mileage “controlling for” year.
- Thus, you can think of the model above as two separate regression models:

For domestic cars, $\text{Origin} = 0 \rightarrow \text{mpg} = \beta_0 + \beta_{\text{Year}}(\text{Year}) + \varepsilon$

For foreign cars, $\text{Origin} = 1 \rightarrow \text{mpg} = (\beta_0 + \beta_{\text{Origin}}) + \beta_{\text{Year}}(\text{Year}) + \varepsilon$

So, β_0 is the intercept for **domestic** cars;

$\beta_0 + \beta_{\text{Origin}}$ is the intercept for **foreign** cars

β_{Origin} is the effect **difference** between domestic and foreign cars



Regression w/Dummy Variable

- {library} Topic
Search keywords
- {ISLR} Bi-Variate Regression Model w/Dummy Variable
`lm.fit <- lm(Sales ~ Price + US,`

Multi-Variate Regression

Multi-Variate Regression

(or Multiple Linear Regression)

- Most regression models include more than one “**predictor**” variable and possibly more than one “**control**” variable
- Predictor and control variables are ALL “**independent variables**”

$$Y = \beta_0 + \beta_1(X_1) + \beta_2(X_2) + \beta_3(X_3) + \dots + \beta_4(X_4) + \beta_5(X_5) + \beta_6(X_6) + \varepsilon$$

Predictors Controls

Independent Variables

- **Interpretation:** β_1 is the **effect size** of X_1 , that is, how much Y changes, when X_1 increases by 1 unit, holding $X_2 \dots X_5$ **constant**. Same for β_2 , etc.
- Predictors and controls work in the exact same way. They are just labels:
 - A **predictor** is a variable we believe will predict Y
 - A **control** is a variable that may affect Y , so we include it for control



Multivariate Regression

- {library} Topic
Search keywords
- {ISLR} Multiple Linear Regression Model

```
lm.fit <- lm(medv ~ lstat + crim + age + chas,  
lm.fit <- lm(medv ~ .,
```
- {coefplot} Coefficient Plot

```
coefplot(lm.fit)
```

ANOVA in Regressions

ANOVA in Regressions

- ANOVA and regression are tightly related because ANOVA tests whether the variance explained by the regression line (or the various variables in the model) is significantly different than the variance of the dependent variable alone.
- As we will see later, ANOVA is very useful when comparing whether one regression model explains more variance than another, so it is a key test when evaluating predictive models.
- The regression's F-test yields a p-value, measuring the overall predictive power of the regression model. What does this measure?

ANOVA in Regressions

- Consider the following 3 **nested** regression models:

Null Model: $Y = \beta_0 + \varepsilon$ (no predictors, just the mean and variance of Y)

Small Model: $Y = \beta_0 + \beta_1 (X_1) + \text{etc.} + \varepsilon$ (a few predictors)

Large Model: $Y = \beta_0 + \beta_1 (X_1) + \dots + \beta_p (X_p) + \text{etc.} + \varepsilon$ (more predictors)

Testing a Single Model

- The **ANOVA F-Test** for a **single model** compares the **variance** of **Y explained** by the regression model (i.e., R^2) against the variance of **Y** explained by the **Null** model

$$F = \frac{R^2 / (p - 1)}{(1 - R^2) / (n - p)}$$

- If **significant** ($p < 0.05$), the model is better at predicting **Y** than the **Null** model.

ANOVA (Nested) Model Comparison

- **anova** (Small, Large) compares the **variance** of **Y explained** by the **Large** model to the variance of **Y** explained by **Small** model

SSE_S = Sum of Squared Errors of the Small model

SSE_L = Sum of Squared Errors of the Large model

df_S = Degrees of freedom of the Small model

df_L = Degrees of freedom of the Large model

$$(ANOVA) F_{(df_S, df_L)} = \frac{\text{SSE reduction per df lost}}{\text{Relative to the Large model}} = \frac{\frac{SSE_S - SSE_L}{df_S - df_L}}{\frac{SSE_L}{df_L}}$$

- If **F** test is **not significant** → retain the **Small model**
- If **F** test is **significant** → retain the **Large model**
- But also **inspect** the selected model for **Multicollinearity**



ANOVA: Comparing Nested Models

- {library} Topic
Search keywords
- anova() Comparing linear models with ANOVA
`anova(lm.null, lm.small, lm.large)`